

Hospital Management System: A Data Analytics Report

1. Project Overview

This project presents a comprehensive analysis of hospital operations using structured healthcare data. The goal is to transform raw transactional data into meaningful insights that support decision making in areas such as revenue optimization, patient care, and operational efficiency.

The solution integrates:

- SQL-based relational database design
- Python (Pandas) for data preprocessing
- Power BI for interactive visualization

2. Database Schema & Data Model

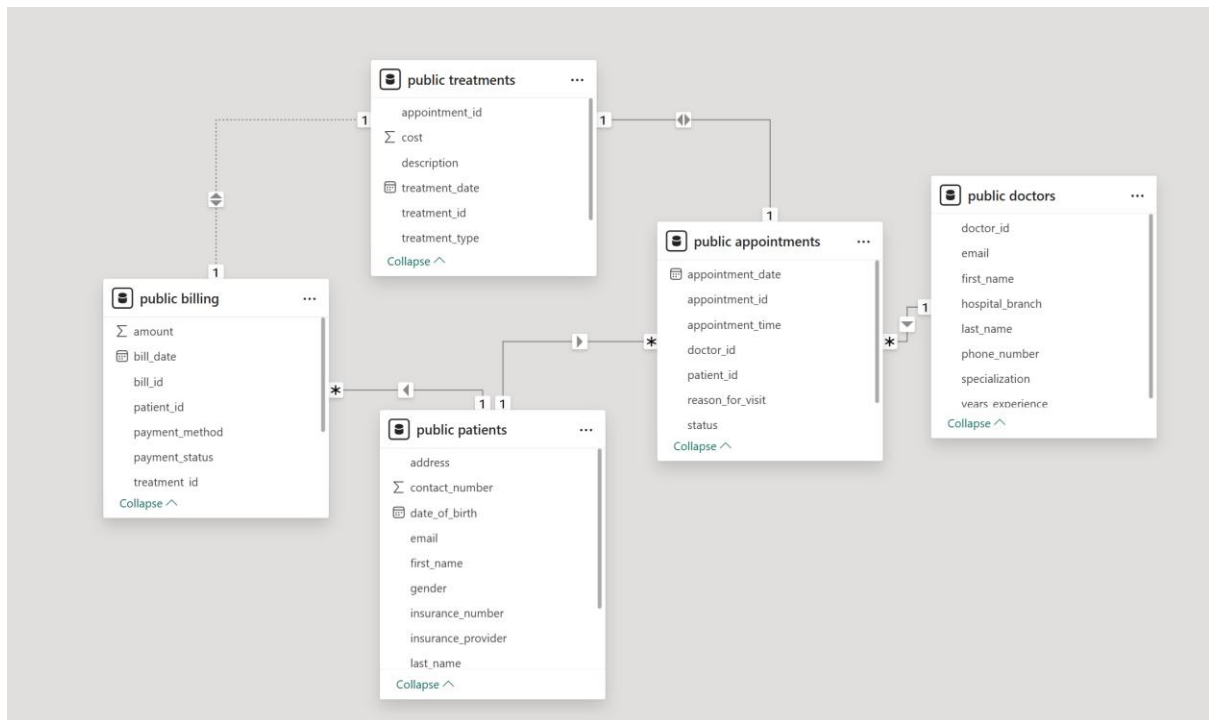
The system is designed using a **relational database model** with strong entity relationships:

Core Tables:

- **Patients** – demographic and insurance details
- **Doctors** – specialization and experience
- **Appointments** – links patients with doctors
- **Treatments** – medical procedures and costs
- **Billing** – financial transactions and payment status

Relationships:

- One **patient** → many **appointments**
- One **doctor** → many **appointments**
- One **appointment** → one **treatment**
- One **treatment** → one **billing record**



Key Insight:

This follows a **star-like schema centered on appointments**, making it ideal for analytical queries and BI tools.

3. Data Processing (ETL Pipeline)

◆ Extraction

Data was imported from multiple CSV sources into a structured environment.

◆ Transformation

Performed using Python (Pandas):

- Handled missing/null values
- Standardized date formats
- Cleaned categorical fields (payment status, specialization, etc.)
- Created derived metrics (e.g., revenue, averages)

```
[1]: import pandas as pd
```

```
[2]: at = pd.read_csv("D:/Projetcs/Hospital Management System/appointments.csv")
      bg = pd.read_csv("D:/Projetcs/Hospital Management System/billing.csv")
      dt = pd.read_csv("D:/Projetcs/Hospital Management System/doctors.csv")
      pt = pd.read_csv("D:/Projetcs/Hospital Management System/patients.csv")
      tt = pd.read_csv("D:/Projetcs/Hospital Management System/treatments.csv")
```

◆ Loading

- Data stored in SQL database
- Connected to Power BI for visualization

```
[23]: from sqlalchemy import create_engine

      username = "postgres"
      password = "Sankalp7*"
      host = "localhost"
      port = "5432"
      database = "hospital"

      engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

      tables = {
          "appointments" : at,
          "billing" : bg,
          "doctors" : dt,
          "patients" : pt,
          "treatments" : tt
      }
      for table_name, df in tables.items():
          df.to_sql(table_name, engine, if_exists="replace", index=False)
          print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")

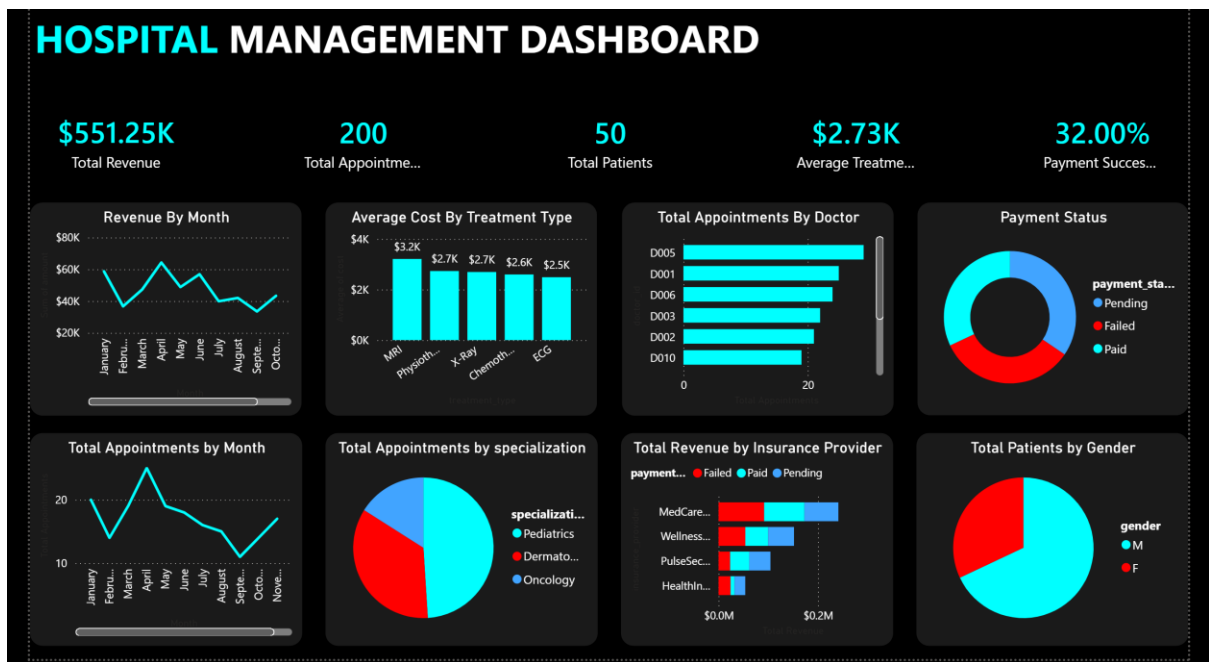
      Data successfully loaded into table 'appointments' in database 'hospital'.
      Data successfully loaded into table 'billing' in database 'hospital'.
      Data successfully loaded into table 'doctors' in database 'hospital'.
      Data successfully loaded into table 'patients' in database 'hospital'.
      Data successfully loaded into table 'treatments' in database 'hospital'.
```

4. Dashboard Overview

The Power BI dashboard provides a centralized view of hospital performance.

Key KPIs:

- **Total Revenue:** \$551.25K
- **Total Appointments:** 200
- **Total Patients:** 50
- **Average Treatment Cost:** \$2.73K
- **Payment Success Rate:** 32%



5. Key Insights & Analysis

➤ Revenue Trends

- Monthly revenue fluctuates significantly
- Peak observed around mid-year
- Drops in later months suggest seasonal or operational inefficiencies

Insight: Revenue instability indicates demand or scheduling issues.

➤ Payment Status (Critical Finding)

- Large proportion of **failed and pending payments**
- Only **32% success rate**

Insight:

This is a major red flag — revenue leakage is high.

Business Impact:

- Cash flow problems
- Inefficient billing system
- Possible insurance processing delays

➤ **Doctor Performance**

- Certain doctors (e.g., D005) handle significantly more appointments
- Uneven workload distribution

Insight:

Resource utilization is imbalanced → burnout risk + inefficiency

➤ **Treatment Cost Analysis**

- MRI is the most expensive (~\$3.2K)
- ECG is the least (~\$2.5K)

Insight:

High-cost treatments contribute disproportionately to revenue.

➤ **Appointment Trends**

- Peak appointments around April
- Decline towards later months

Insight:

Demand variability → potential scheduling inefficiencies

➤ **Specialization Distribution**

- Pediatrics dominates
- Oncology has the least share

Insight:

Hospital demand is skewed → staffing and resource allocation should match demand.

➤ **Insurance Revenue Analysis**

- Majority revenue comes from specific providers (e.g., Medicare-like providers)
- Payment status varies across providers

Insight:

Some insurers may delay or fail payments more frequently.

➤ **Patient Demographics**

- Male patients significantly higher than female

Insight:

Could indicate demographic bias or dataset imbalance.

6. Key Problems Identified

 1. Low Payment Success Rate (32%)

- Major revenue leakage

 2. Uneven Doctor Workload

- Inefficient resource allocation

 3. Revenue Instability

- Seasonal or operational issues

 4. High Dependency on Few Insurance Providers

- Risk concentration

7. Recommendations

 Financial Improvements

- Implement stricter billing validation
- Automate payment follow-ups
- Improve insurance claim processing

 Operational Efficiency

- Balance doctor workload using scheduling optimization
- Forecast patient demand using time series models (fits your thesis 🧠)

 Strategic Decisions

- Expand high-demand specializations (e.g., Pediatrics)
- Improve marketing or outreach in low-demand areas

 Data Improvements

- Track payment delays and reasons
- Add patient satisfaction metrics

8. Tools & Technologies Used

- **SQL** → Database design & querying

```

60
61 --11. Find top 2 patients (per doctor) with most visits
62 SELECT * FROM (SELECT p.patient_id, d.doctor_id, COUNT(p.patient_id), ROW_NUMBER() OVER (PARTITION BY d.doctor_id ORDER
63 JOIN appointments a ON p.patient_id = a.patient_id
64 JOIN doctors d ON d.doctor_id = a.doctor_id
65 GROUP BY d.doctor_id, p.patient_id)
66 WHERE rn <=2;
67
68 --12. Identify patients with highest total billing amount using CTE
69 WITH total_revenue AS
70 (SELECT p.patient_id, SUM(amount) as total_bill FROM patients p
71 JOIN billing b ON p.patient_id = b.patient_id
72 GROUP BY p.patient_id
73 ORDER BY total_bill DESC)
74
75 SELECT patient_id, ROUND(total_bill::numeric,2) AS total_amount, RANK() OVER (ORDER BY total_bill DESC) FROM total_revenue

```

Data Output Messages Notifications

Showing rows: 1 to 48 Page No: 1 of 1

	patient_id text	total_amount numeric	rank bigint
1	P012	30053.08	1
2	P049	23554.06	2
3	P016	22967.94	3
4	P036	21583.56	4
5	P025	19513.17	5
6	P005	18609.91	6
7	P035	18407.42	7

- **Python (Pandas, NumPy)** → Data cleaning & transformation

1	P002	Emily	Smith	F	12-10-1984	8228188767	321 Maple Dr	15-01-2022	PulseSecure	INS354079	emily.smith@mail.cor
2	P003	Laura	Jones	M	21-08-1977	8397029847	321 Maple Dr	07-02-2022	PulseSecure	INS650929	laura.jones@mail.cor
3	P004	Michael	Johnson	NaN	20-02-1981	9019443432	123 Elm St	02-03-2021	HealthIndia	INS789944	michael.johnson@mail.cor
4	P005	David	Wilson	M	23-06-1960	7734463155	123 Elm St	29-09-2021	MedCare Plus	INS788105	david.wilson@mail.cor

```

[9]: pt['date_of_birth'] = pd.to_datetime(pt['date_of_birth'], format = '%d-%m-%Y')
    pt['registration_date'] = pd.to_datetime(pt['registration_date'], format = '%d-%m-%Y')

[10]: tt.head()

[10]: treatment_id appointment_id treatment_type description cost treatment_date
0 T001 A001 Chemotherapy Basic screening 3941.97 09-08-2023
1 T002 A002 MRI Advanced protocol 4158.44 09-06-2023
2 T003 A003 MRI Standard procedure 3731.55 28-06-2023
3 T004 A004 MRI Basic screening 4799.86 01-09-2023
4 T005 A005 ECG Standard procedure 582.05 06-07-2023

[11]: tt['treatment_date'] = pd.to_datetime(tt['treatment_date'], format = '%d-%m-%Y')

```

- **Power BI** → Data visualization & dashboarding

9. Conclusion

This project demonstrates how healthcare data can be leveraged to uncover operational inefficiencies and improve decision-making.

The analysis highlights critical issues such as:

- Low payment success rate
- Revenue inconsistency
- Resource imbalance

By implementing the recommended solutions, hospitals can significantly improve both **financial performance and patient service quality**.